Lab 3 – Component Modelling & Architectural Pattern Selection

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Architecture Selection: We chose Layered Architecture for the Self-Service Coffee Kiosk System.

1. Architectural Choice

Layered (n-tier) architecture with three layers: **Presentation** (Touchscreen UI), **Business** (Order Manager, Payment Service) and **Data & Device Access** (Receipt Printer, Menu Database). Each layer calls only the layer below it through a defined interface (IMenu, IOrder, IPayment, IReceipt, IMenuData).

2. Two Reasons

(a) The workflow is one fixed, linear transaction on a single device – choose drink and size, pay by credit card, print the receipt – with only 3 drinks × 2 sizes. This maps directly onto layers, and each of the five components has one responsibility, so the UI, payment provider or printer model can be replaced behind its interface without touching the rest.

(b) The alternatives cost more than they return at this scale. Microservices would add network hops, service discovery and deployment overhead for one kiosk. Client-Server would make the kiosk depend on a central server, a single point of failure that stops sales if it goes down. Layered runs entirely on the kiosk (only card authorisation needs the network) and is simple to build, test and maintain.

3. Security Advantage

Strict layering gives one controlled path into the system. The touchscreen UI cannot reach the payment service or the database directly; every request passes through the Order Manager, which validates it (only valid drink/size combinations, prices looked up from the database rather than supplied by the UI). Card data is isolated in the Payment Service, sent only over TLS to the payment gateway and never stored; the receipt shows only masked digits. This keeps PCI-DSS scope to a single component.

4. Performance Benefit

All layers run in-process on the kiosk, so internal interface calls are local method calls with no network latency, and menu and pricing data come from a local embedded database (cacheable by the Order Manager). Touch responses stay instant and the only remote call per order is one payment authorisation. Trade-off: layering adds slight call overhead and cannot scale layers independently, which is acceptable for a single kiosk.

Style	Fit for the kiosk	Verdict
Layered	Clear separation of concerns, simple, runs locally; weaker per-component scaling (not needed here)	**Selected**
Microservices	Independent scaling and fault isolation, but high operational complexity and latency	Over-engineered
Client-Server	Central control and easy consistency, but single point of failure and bottleneck	Rejected